We are using the concept of Lateral Surface Area (LSA) and Total Surface Area (TSA) in real life situations. For instance a room can be cuboidal in shape that has different length, breadth and height. If we require to find areas of only the walls of a room, avoiding floor and ceiling then we can use LSA. However if we want to find the surface area of the whole room then we have to calculate the TSA.

If the length, breadth and height of a cuboid are l, b and h respectively. Then

(i) Total Surface Area $= 2\,(lb + bh + lh)$ sq.units.

(ii) Lateral Surface Area $= 2\,(l+b)h$ sq.units.

Note

- The top and bottom area in a cuboid is independent of height. The total area of top and bottom is $2lb$. Hence LSA is obtained by removing $2lb$ from $2(lb+bh+lh)$.
- The units of length, breadth and height should be same while calculating surface area of the cuboid.

Example 7.4 Find the TSA and LSA of a cuboid whose length, breadth and height are 7.5 m, 3 m and 5 m respectively.

Solution

Given the dimensions of the cuboid;

that is length $(l) = 7.5\ m$, breadth $(b) = 3\ m$ and height $(h) = 5\ m$.

$$\text{TSA} = 2(lb + bh + lh)$$

$$= 2\left[(7.5\times 3) + (3\times 5) + (7.5\times 5)\right]$$

$$= 2(22.5 + 15 + 37.5)$$

$$= 2\times 75$$

$$= 150\ m^2$$

$$\text{LSA} = 2(l + b)\times h$$

$$= 2(7.5 + 3)\times 5$$

$$= 2\times 10.5\times 5$$

$$= 105\ m^2$$

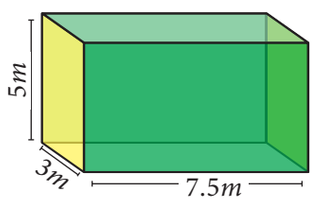


Fig. 7.11

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